Hurwitz counts in the finite-group TopFT

Clarifications of the CohFT note, pp. 5–9

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This consolidates the finite-group example in *TopFT and CohFT*, §1.3, pp. 5–6, and the pairing, handle element, and Burnside calculation in §1.4, pp. 8–9. Throughout, g is the genus of the *target*; h will denote the genus of a *connected source*.

1. The coefficient and the count for a general finite group

Let G be finite, $V = Z(\mathbb{C}[G])$, and $C_K = \sum_{x\in K} x$ for a conjugacy class $K \subseteq G$. The conventions of the note are

$$\alpha = \sum_{a,b\in G} [a,b], \qquad [a,b] = aba^{-1}b^{-1}, \qquad \omega_{g,n,0}(v_1,\dots,v_n) = \frac{1}{|G|}[1_G](v_1\cdots v_n\alpha^g).$$

What does $[1_G]$ mean? It extracts the coefficient of the identity in the *group-element basis* of $\mathbb{C}[G]$:

$$z = \sum_{x\in G} a_x x \quad\Longrightarrow\quad [1_G](z) = a_{1_G}.$$

For a central element, write instead $z = \sum_K b_K C_K$. Then $a_x = b_K$ for $x \in K$, and in particular

$$\boxed{[1_G](z) = b_{\{1_G\}}, \qquad C_{\{1_G\}} = 1_G.}$$

Thus the two interpretations in the question give *the same answer* in the unnormalized class-sum basis used in the note. There is no extra class-size factor at this step. Multiplication still takes place in the group algebra; multiplying class sums is not coordinatewise multiplication.

What is counted? For class-sum inputs, expanding the product gives

$$\omega_{g,n,0}(C_{K_1},\dots,C_{K_n}) = \frac{\#\mathcal{T}_G}{|G|}, \qquad \mathcal{T}_G = \left\{ (a_1,b_1,\dots,a_g,b_g,c_1,\dots,c_n) \;\middle|\; \begin{array}{l} a_i,b_i\in G,\ c_j\in K_j,\\ \prod_{i=1}^{g}[a_i,b_i]\prod_{j=1}^{n} c_j = 1_G \end{array} \right\}. \tag{1}$$

Indeed, every choice contributes one group element, and $[1_G]$ retains exactly the choices satisfying the displayed relation. The class sums are central, so the order of the handle block and the class-sum block does not change the result.

Fix a smooth connected complex curve Y of genus g and distinct points $q_1,\dots,q_n$. The relation in (1) is the standard presentation of $\pi_1(Y\setminus\{q_j\})$, with c_j the local monodromy around q_j. Thus $\mathcal{T}_G$ parametrizes homomorphisms with prescribed local conjugacy classes. Simultaneous conjugation by G changes the trivialization at the base point. By orbit–stabilizer,

$$\frac{\#\mathcal{T}_G}{|G|} = \sum_{[\varphi]} \frac{1}{|C_G(\operatorname{im}\varphi)|} = \sum_{[P]} \frac{1}{|\operatorname{Aut}_G(P)|}.$$

The last sum is over principal G-bundles on the punctured curve with those local monodromies, and automorphisms are over that curve and G-equivariant. Their total spaces need not be connected; connectedness would require φ to be surjective. For arbitrary inputs $v_j = \sum_K b_{j,K} C_K$, multilinearity gives the corresponding linear combination of these counts, with weight $\prod_j b_{j,K_j}$.

2. For $G = S_d$: which covers, and which source genus?

A partition $\mu = (\mu_1, \dots, \mu_\ell) \vdash d$ describes the cycle lengths of a permutation. Write $K_\mu \subset S_d$ for this conjugacy class and $C_\mu = \sum_{\sigma \in K_\mu} \sigma$. Define

$$H_g^\bullet(\mu_1, \dots, \mu_n) := \omega_{g,n,0}(C_{\mu_1}, \dots, C_{\mu_n});$$

here each μ_j is a whole partition, and $\bullet$ means that disconnected sources are allowed.

This counts degree-d maps $f : X \to Y$, with Y and the q_j fixed, such that the ramification profile over q_j is μ_j and f **is unramified over** $Y \setminus \{q_1, \dots, q_n\}$**.** Each isomorphism class has weight $1/|\operatorname{Aut}_Y(f)|$. The source is a smooth compact curve, possibly disconnected; every component maps nonconstantly to Y. Neither sheets nor points in the fibers are labeled.

The monodromy construction explains the count: a homomorphism to S_d gives a cover of the punctured curve on d letters; a cycle of length m is filled in by the local map $z \mapsto z^m$. Relabeling the letters is simultaneous conjugation. Its stabilizer is $\operatorname{Aut}_Y(f)$, so division by $d!$ gives precisely the weights above. Equivalently, the principal S_d-bundle is the bundle of labelings of the d sheets; the associated cover has degree d, not $d!$.

Let H_g° denote the count with *connected* source. In (1), it is obtained by requiring

$$\langle a_1, b_1, \dots, a_g, b_g, c_1, \dots, c_n \rangle \quad \text{to act transitively on } \{1, \dots, d\}.$$

Transitivity is enough; the monodromy need not be all of S_d.

Riemann–Hurwitz determines h**.** Let $\ell(\mu)$ be the number of parts of a partition. Above a point with profile μ, the contribution to the ramification divisor is

$$\sum_{x \mapsto q} (e_x - 1) = \sum_{k=1}^{\ell(\mu)} (\mu_k - 1) = d - \ell(\mu).$$

Consequently, for a connected source of genus h,

$$\boxed{2h - 2 = d(2g-2) + \sum_{j=1}^{n} (d - \ell(\mu_j)), \qquad h = 1 + d(g-1) + \frac{1}{2} \sum_{j=1}^{n} (d - \ell(\mu_j)).} \tag{2}$$

There is no independent choice of h in this count. If the right-hand side for h is not a nonnegative integer, there is no connected cover. The Riemann–Hurwitz condition is necessary, but by itself does not guarantee that a transitive monodromy tuple exists.

For the disconnected count, use Euler characteristic. If $X = \bigsqcup_{a=1}^{c} X_a$ and X_a has genus h_a, then

$$\chi_{\text{top}}(X) = \sum_{a=1}^{c} (2 - 2h_a) = d(2-2g) - \sum_{j=1}^{n} (d - \ell(\mu_j)).$$

The data fix this Euler characteristic, not c or the individual h_a. The number obtained by the formula for h in (2) is now $1 - c + \sum_a h_a$ (the arithmetic genus $1 - \chi(\mathcal{O}_X)$), which can be negative. The picture labeled "genus h" on p. 5 of the original note should therefore be read as the connected-source case.

Small check. For $d = 2$, $g = 0$, and profiles $(2), (2)$, there is one monodromy tuple, giving $H_0^\bullet((2),(2)) = 1/2$; the cover is connected of genus 0. With no branch points, $H_0^\bullet(\varnothing) = 1/2$ instead counts $\mathbb{P}^1 \sqcup \mathbb{P}^1 \to \mathbb{P}^1$, whose two sheets can be interchanged; its arithmetic genus is -1.

3. Extra branch points and the single/double Hurwitz conventions

For $H_g^\bullet(\mu_1,\dots,\mu_n)$ as just defined, the number of *other* branch points is **zero**. Riemann–Hurwitz computes the source Euler characteristic from this complete branch data.

The familiar alternative is to prescribe a connected source genus h and allow *additional simple branching*. For $d \geq 2$, set $\tau = (2, 1^{d-2})$. If the extra branching is simple and lies over r distinct additional fixed points of Y, then

$$\boxed{r = 2h - 2 - d(2g-2) - \sum_{j=1}^{n}(d - \ell(\mu_j)).} \tag{3}$$

Each transposition contributes one to the ramification divisor. Thus the corresponding possibly disconnected count is

$$H_g^\bullet(\mu_1,\dots,\mu_n,\underbrace{\tau,\dots,\tau}_{r}) = \omega_{g,n+r,0}(C_{\mu_1},\dots,C_{\mu_n},\underbrace{C_\tau,\dots,C_\tau}_{r}).$$

For the connected count impose transitivity as before. A negative value of r rules out this problem. Without the assumption of simple branching, the difference in (3) specifies only the *total remaining ramification*, not the number of remaining branch points. Branch points are on Y; ramification points are on X.

For target $Y = \mathbb{P}^1$, the standard terminology is:

- **Single Hurwitz numbers:** one distinguished profile μ and $r = 2h - 2 + d + \ell(\mu)$ additional simple branch points. The connected count $H_0^\circ(\mu, \tau, \dots, \tau)$ is the one appearing in the usual ELSV formula.
- **Double Hurwitz numbers:** two distinguished profiles μ, ν and $r = 2h - 2 + \ell(\mu) + \ell(\nu)$ additional simple branch points.

Thus the TopFT packages *general Hurwitz numbers with several prescribed profiles*. It includes these single and double problems by inserting the required transposition class sums. An n-input correlator does not silently add simple branching. Our normalization uses fixed branch points, unlabelled preimages, and weight $1/|\operatorname{Aut}_Y(f)|$; no $1/r!$ is included. This is the connected single-Hurwitz convention in Liu, §2.1, pp. 4–5; her Theorem 2, p. 8, is the ELSV formula.

4. What does a character evaluated at a partition mean?

For a representation $\rho : G \to \mathrm{GL}(V_\rho)$, $\chi_\rho(x) = \mathrm{tr}(\rho(x))$. Since trace is invariant under conjugation, χ_ρ is constant on each conjugacy class. Consequently

$$\boxed{\chi_\rho(\mu) := \chi_\rho(\sigma) \quad \text{for any permutation } \sigma \text{ of cycle type } \mu.}$$

The partition is simply the label for the conjugacy class, not itself a group element. Irreducible representations of S_d are *also* indexed by partitions: if $\rho = \rho_\lambda$, then $\chi^\lambda(\mu)$ is the character-table entry with representation label λ and conjugacy-class label μ.

For example, in the two-dimensional standard representation of S_3, the character is the number of fixed letters minus one. Hence

$$\chi^{(2,1)}(1^3) = 2, \qquad \chi^{(2,1)}(2,1) = 0, \qquad \chi^{(2,1)}(3) = -1.$$

By contrast, $\rho(C_\mu) = \sum_{\sigma \in K_\mu} \rho(\sigma)$ has trace $|K_\mu|\chi_\rho(\mu)$; its scalar action on an irreducible representation is $|K_\mu|\chi_\rho(\mu)/\dim\rho$.

5. The pairing, the handle, and Burnside's formula together

Here is the calculation from pp. 8–9, with its normalizations collected in one place. Put $\varepsilon(z) = [1_G](z)/|G|$. A product xy equals 1_G exactly when $y = x^{-1}$, so

$$\eta(C_K, C_L) = \varepsilon(C_K C_L) = \delta_{L,K^{-1}} \frac{|K|}{|G|}, \qquad \alpha = \sum_K \frac{|G|}{|K|} C_K C_{K^{-1}}.$$

The second formula uses the inverse pairing to glue a handle. It equals $\sum_{a,b}[a,b]$: for fixed $b \in K$, each conjugate of b occurs $|G|/|K|$ times among the aba^{-1}, and summing b^{-1} over K gives $C_{K^{-1}}$.

Let p_ρ be the primitive central idempotent projecting onto the ρ-isotypic summand, and put $d_\rho = \dim V_\rho$ and $t_\rho = \varepsilon(p_\rho)$. Then

$$p_\rho p_\sigma = \delta_{\rho\sigma} p_\rho, \qquad 1_G = \sum_\rho p_\rho, \qquad t_\rho = \left(\frac{d_\rho}{|G|}\right)^2.$$

For the last identity, left multiplication L_z on $\mathbb{C}[G]$ has trace $|G|[1_G](z)$. Since $\mathbb{C}[G] \cong \bigoplus_\rho V_\rho^{\oplus d_\rho}$, L_{p_ρ} has trace d_ρ^2. Applying the normalization of ε gives $t_\rho = d_\rho^2/|G|^2$.

The pairing is diagonal in this basis, so the handle element and the class sums become

$$\alpha = \sum_\rho t_\rho^{-1} p_\rho, \qquad C_K = \sum_\rho \lambda_\rho(K) p_\rho, \qquad \lambda_\rho(K) = \frac{|K|\chi_\rho(K)}{d_\rho}.$$

Indeed, Schur's lemma makes the action of C_K scalar, and taking traces gives $d_\rho \lambda_\rho(K) = |K|\chi_\rho(K)$. Multiplying these expansions and applying ε now gives

$$\boxed{\omega_{g,n,0}(C_{K_1}, \dots, C_{K_n}) = \sum_{\rho \in \mathrm{Irr}(G)} \left(\frac{d_\rho}{|G|}\right)^{2-2g} \prod_{j=1}^n \frac{|K_j|\chi_\rho(K_j)}{d_\rho}.} \tag{4}$$

Each handle contributes t_ρ^{-1} and the final counit contributes t_ρ, explaining the exponent $1-g$ of t_ρ. In particular, coefficient extraction in the idempotent basis is *not* just taking one coordinate: if $z = \sum_\rho z_\rho p_\rho$, then $[1_G](z) = \sum_\rho z_\rho d_\rho^2/|G|$.

For S_d, if $\mu = (1^{m_1} 2^{m_2} \cdots)$, let $z_\mu = \prod_{k \geq 1} k^{m_k} m_k!$. Its conjugacy class has size $|K_\mu| = d!/z_\mu$: cycles may be rotated, and cycles of equal length may be permuted in the centralizer. Thus (4) reads

$$\boxed{H_g^\bullet(\mu_1, \dots, \mu_n) = \sum_{\rho \in \mathrm{Irr}(S_d)} \left(\frac{d_\rho}{d!}\right)^{2-2g} \prod_{j=1}^n \frac{(d!/z_{\mu_j})\chi_\rho(\mu_j)}{d_\rho}.}$$

The notation $|C_\mu|$ in the original note means $|K_\mu|$ here. This formula counts *possibly disconnected* covers. A connected answer requires the transitivity restriction; it is not obtained by restricting the irreducible representations in the sum.

Source locations. *TopFT and CohFT* (updated file `2026-11-01-topft-cohft.pdf`), §1.3, pp. 5–6; §1.4, pp. 8–9. The calculation above expands its unnumbered "Recovering Burnside's formula" on p. 9. For the general finite-group interpretation and Frobenius normalization, see C. Teleman, *Five lectures on Topological Field Theory*, §1.3, pp. 1–2, and §§1.8–1.10, p. 3. For the single-Hurwitz convention and ELSV comparison, see C.-C. M. Liu, *Lectures on the ELSV formula*, §2.1, pp. 4–6, and Theorem 2, p. 8.